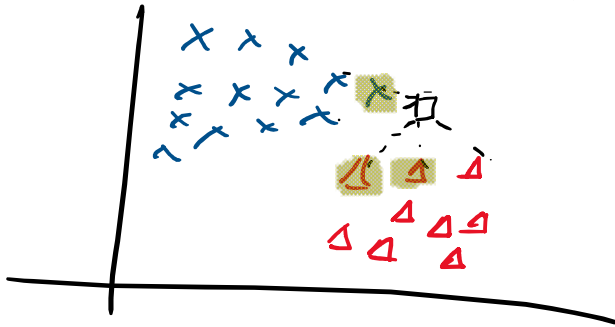


$\Rightarrow$ Non-parametric Methods :

$\hookrightarrow$ K-Nearest Neighbors :



$k=3$

- i) Minkowski Distance
- ii) Euclidean Distance
- iii) Manhattan Distance
(Taxicab Distance)

i) Minkowski Distance :

$$D_p(x_0, x_1) = \sqrt[p]{\sum_{j=1}^n |x_{0,j} - x_{1,j}|^p}$$

ii) Euclidean Distance :

$$D(x_0, x_1) = \sqrt{\sum_{j=1}^n (x_{0,j} - x_{1,j})^2}$$

iii) Manhattan Distance :

$$D_1(x_0, x_1) = \sum_{j=1}^n |x_{0,j} - x_{1,j}|$$

$$D(x_0, x_i) = \sum_{j=1}^n |x_{0,j} - x_{i,j}|$$

→ Sorting & Neighborhood Segmentation:

$$d(x_{(1)}) \leq d(x_{(2)}) \leq \dots \leq d(x_{(k)}) \leq \dots \leq d(x_{(n)})$$

Neighborhood Set:

$$N_k(x_0) = \{x_{(1)}, x_{(2)}, \dots, x_{(k)}\}$$

⇒ Inference: Majority Voting:

$$\hat{y} = \arg \max_{c \in [1, \dots, C]} \sum_{x_i \in N_k(x_0)} I(y_i = c)$$

Example: $k=3$

x_0	BMI	x_i	Age	Sugar	Distance	K_{rank}	
33.6		50		1	14.14	(1)	
26.6		30		0	19.72	4	
23.4		40		0	20.20	5	
43.1		67		0	27.00	8	
35.3		23		1	18.92	(3)	
35.9		67		1	28.08	9	

Test Example

y_0 BMI = 43.6

y_1 Age = 40

!!!

35.9	67	1	28.08
25.7	46	0	18.88
23.3	29	0	22.09
31.	56	1	20.37

$\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$
 9
 (2)
 7
 6
 ∴ Person has sym.

⇒ Euclidean : i) $\sqrt{(y_{0,i} - x_{0,i})^2 + (y_{1,i} - x_{1,i})^2}$

$$= \sqrt{(43.6 - 33.6)^2 + (40 - 50)^2}$$

$$= 14.14$$

$$\text{ii) } \sqrt{(y_{0,3} - x_{0,3})^2 + (y_{1,3} - x_{1,3})^2}$$

$$\sqrt{(43.6 - 23.4)^2 + (40 - 40)^2}$$

$$= 20.20$$

⇒ Determining the value of k:

i) Take the set of N of values S select that as k .

ii) Elbow Determination:



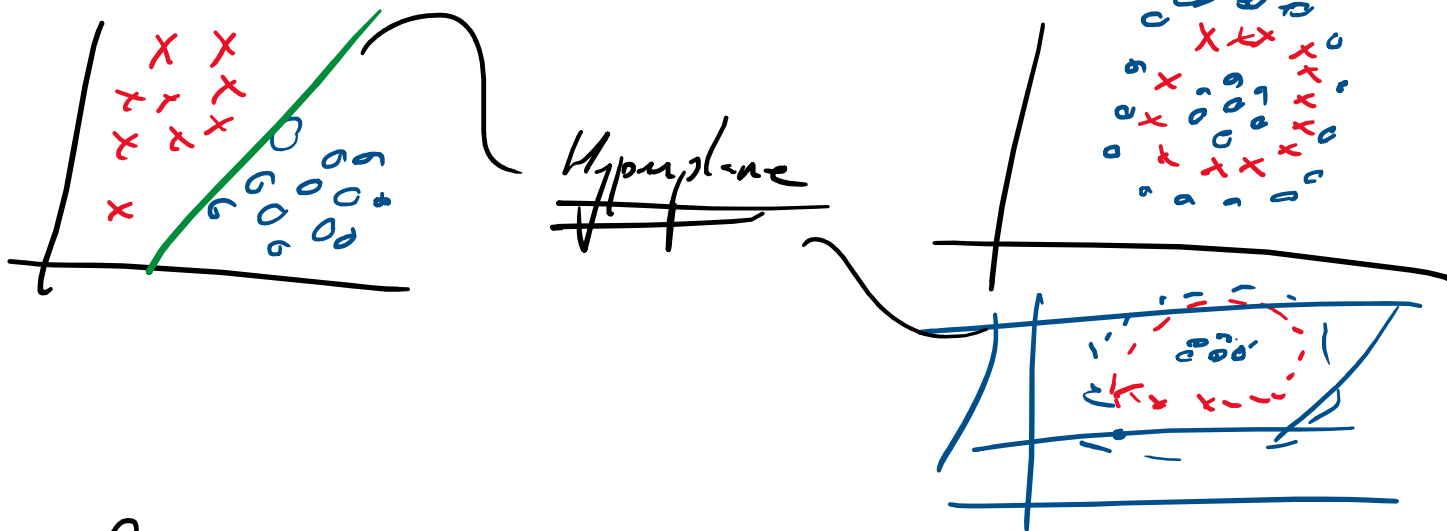
iii) No. of classes + 1 as the value of $\underline{k}$ (Even)
 Classes + 2 (Odd)

⇒ Example 2: $k = 5$

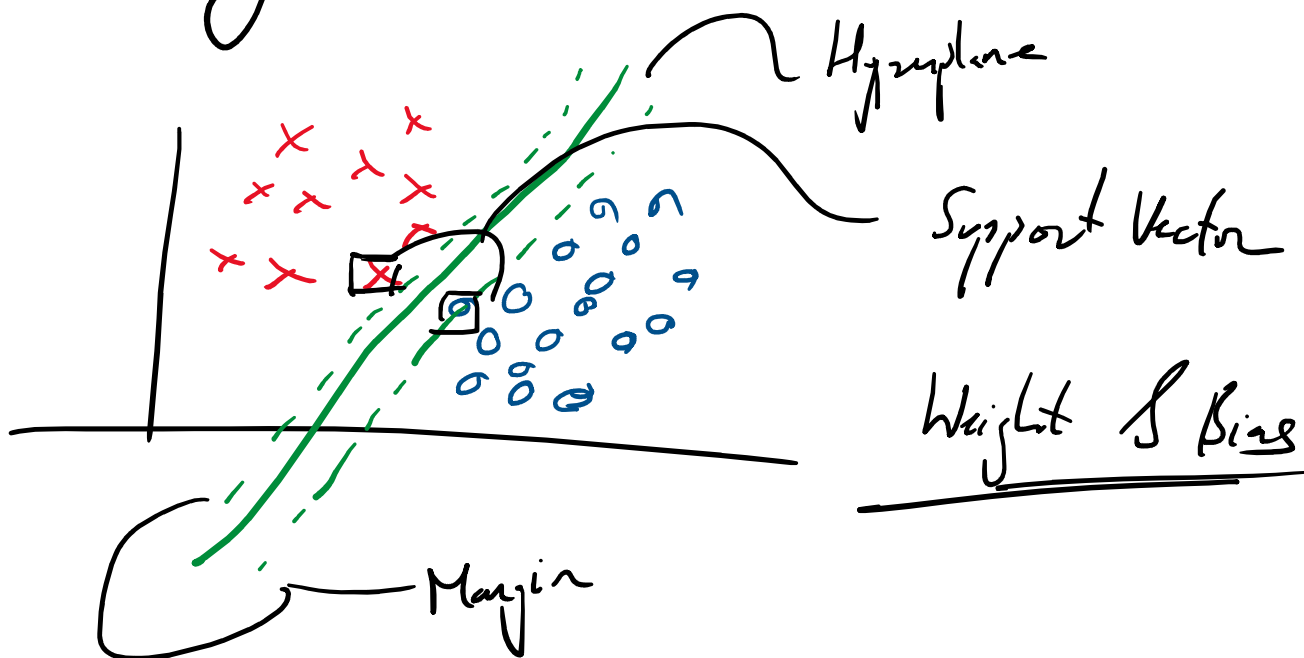
Height	Weight	Class	Distance	Rank
169	58	Underweight	1.4	1
170	55	Normal	2	2
173	57	Normal	3	3
174	56	Underweight	4.1	4
167	51	Underweight	6.7	5
173	64	Normal	7.6	6
172	65	Normal	8.2	7
182	62	Underweight	13	8
174	69	Normal	13.4	9

Height
170
Weight
57

$\Rightarrow$ Support Vector Machines:



Svm works with linearly separable & non-linearly separable data.



Hypersurface: $|W^T x + b = 0|$

Hyperplane: $W^T x + b = 0$

→ Define Margin Geometrically:

$$\text{Distance} = \frac{(W^T x + b)}{\|w\|}$$

$$y_i (W^T x + b) \geq 1, \forall i = 1, \dots, N$$

$$\text{Margin width} = \frac{1}{\|w\|} + \frac{1}{\|w\|} = \boxed{\frac{2}{\|w\|}}$$

Step 1:

⇒ Dual Function (Lagrangian Dual)

$$q(\vec{a}) = \sum_{i=1}^N a_i - \frac{1}{2} \sum_{i=1, j=1}^N a_i a_j y_i y_j (\vec{x}_i \cdot \vec{x}_j)$$

Subject to

$$\sum_{i=1}^N a_i y_i = 0$$

for $i = 1, 2, \dots, N$

$$a_1 + a_2 + a_3 + \dots + a_n = 0$$

$$a_1 + a_2 + a_3 \dots a_n = 0$$

Step 2:

Compute

$$\vec{w} = \sum_{i=1}^N a_i y_i \vec{x}_i$$

Compute

$$b = \frac{1}{2} \left(\min_{i: y_i = +1} (\vec{w} \cdot \vec{x}_i) + \max_{i: y_i = -1} (\vec{w} \cdot \vec{x}_i) \right)$$

$\Rightarrow$ SVM Classifier

$$f(\vec{x}) = \vec{w} \cdot \vec{x} - b$$

Example: $N=2$

$$\vec{x}_1 = (2, 1) \quad \vec{x}_2 = (4, 3)$$

$$y_1 = +1 \quad y_2 = -1$$

x_1	x_2	Class
2	1	+1
4	3	-1

$$\vec{a} = (a_1, a_2)$$

Subject to conditions

$$a_1 - a_2 = 0 \quad \text{means} \quad a_1 = a_2$$

$$a_1 > 0, \quad a_2 > 0$$

Step 1:

$$\begin{aligned}
 \phi(\vec{a}) &= \sum_{i=1}^N a_i - \frac{1}{2} \sum_{i=1, j=1}^N a_i a_j y_i y_j (\vec{x}_i \cdot \vec{x}_j) \\
 &= (a_1 + a_2) - \frac{1}{2} \left[a_1 a_1 y_1 y_1 (\vec{x}_1 \cdot \vec{x}_1) + \right. \\
 &\quad a_1 a_2 y_1 y_2 (\vec{x}_1 \cdot \vec{x}_2) + \\
 &\quad a_2 a_1 y_2 y_1 (\vec{x}_2 \cdot \vec{x}_1) + \\
 &\quad \left. a_2 a_2 y_2 y_2 (\vec{x}_2 \cdot \vec{x}_2) \right] \\
 &= (a_1 + a_2) - \frac{1}{2} \left[a_1^2 (+1)(+1)(2 \times 2 + 1 \times 1) + \right. \\
 &\quad a_1 a_2 (+1)(-1)(2 \times 4 + 1 \times 3) + \\
 &\quad a_2 a_1 (-1)(+1)(4 \times 2 + 3 \times 1) + \\
 &\quad \left. a_2^2 (-1)(-1)(4 \times 4 + 3 \times 3) \right] \\
 &= (a_1 + a_2) - \frac{1}{2} [5a_1^2 - 22a_1 a_2 + 25a_2^2]
 \end{aligned}$$

Find values of a_1 & a_2 which maximizes

$$\phi(\vec{a}) = (a_1 + a_2) - \frac{1}{2} [5a_1^2 - 22a_1 a_2 + 25a_2^2]$$

$$\phi(\vec{a}) = 2a_1 - 4a_1^2$$

1 1

$$|a_1| - a_1 - \dots$$

To be maximum we must have

$$db/da_1 = 2 - 8a_1 = 0$$

$$a_1 = \frac{1}{4}$$

$$a_2 = \frac{1}{4}$$

Step 2:

$$\vec{w} = \sum_{i=1}^N a_i y_i \vec{x}_i$$

$$= a_1 y_1 \vec{x}_1 + a_2 y_2 \vec{x}_2$$

$$= \frac{1}{4} (+1) (2, 1) + \frac{1}{4} (-1) (4, 3)$$

$$= \frac{1}{4} (-2, -2)$$

$$\vec{w} = \left(-\frac{1}{2}, -\frac{1}{2} \right)$$

$$b = \frac{1}{2} \left(\min(\vec{w} \cdot \vec{x}_i) + \max(\vec{w} \cdot \vec{x}_i) \right)$$

$$= \frac{1}{2} \left((\vec{w} \cdot \vec{x}_1) + (\vec{w} \cdot \vec{x}_2) \right)$$

$$= \frac{1}{2} \left(\left(-\frac{1}{2}x_2 - \frac{1}{2}x_1 \right) + \left(-\frac{1}{2}x_4 - \frac{1}{2}x_3 \right) \right)$$

$$b = \frac{1}{2} \left(-\frac{10}{2} \right) = \boxed{-\frac{5}{2}}$$

Hg) 3:

$$f(\vec{x}) = \vec{w} \cdot \vec{x} - b$$

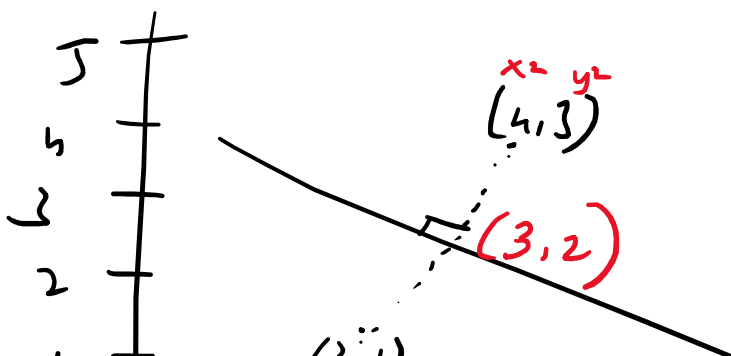
$$= \left(-\frac{1}{2}, -\frac{1}{2} \right) \cdot (x_1, x_2) - \left(-\frac{5}{2} \right)$$

$$= -\frac{1}{2}x_1 - \frac{1}{2}x_2 + \frac{5}{2}$$

$$= -\frac{1}{2} (x_1 + x_2 - 5)$$

$$f'(\vec{x}) = 0$$

$$\therefore f(\vec{x}) \approx \boxed{x_1 + x_2 - 5 = 0}$$



$$\frac{x_1 + x_2}{2} = \frac{2+4}{2} = 3$$

$$\frac{y_1 + y_2}{2} = 2$$

$$x_1 + x_2 - 5 = 0$$

